CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (CL) Examination

November

CL 213 Building Construction

Date: 30. 11. 2013 | Time: 01:30 a.m. To 04:30 p.m. | Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a)		Write requirements of good Stair.	03
(b)		Sketch quarter turn staircase. Also, state in which case it is provided.	04
Q - 2 (a)		Sketch wall section showing various components for brick masonry work.	05
(b)		Sketch elevation for Flemish bond brick masonry wall.	03
(c)		Sketch elevation of Ashlar chamfered stone masonry wall.	03
(d)	i.	Which paint is used for water proofing?	03
	ii.	Name the paint by which the painted surface can be washed with water.	
	iii.	Write the name of water paint in which oil is not used.	

OR

Q - 2 (a)		Explain English bond with plan and elevation for 1 brick thick wall.	05
(b)		What is composite masonry? Write types of it.	03
(c)		What is frog? State use of it. Write nominal size of traditional brick.	03
(d)	i.	Which paint is used for protection of iron and steel under water?	03
	ii.	Which paint is used for electric and telegraphic poles and towers?	
	iii.	Which paint can be mixed with water instead of solvent?	

Q - 3 (a)	i.	Which type of door is suggested at libraries, museums, banks etc?	04
	ii.	Which type of door is suitable where opening is large but there is no enough space to accommodate leafed shutter?	
	iii.	Which type of window is provided for the sole purpose of light and vision only?	
	iv.	Which window provides extra space in room and improve the appearance also.	

	(b)	Sketch casement window with label.	04
	(c)	Sketch swing door showing all components.	04
	(d)	Write the names only for component parts of scaffolding.	02

OR

Q - 3	(a)	What are the types and size of door frame?	04
	(b)	Sketch single leaf two paneled door with elevation, horizontal and vertical section details. Write nominal size of door. Also write suitability of this type of door.	08
	(c)	What are the types of shores? Which shore is used when the distance between two buildings is 12.5m?	02

SECTION - II

Q - 4	(a)	What are the reasons of adopting/selecting raft foundations over isolated or combine foundation for a building?	03
	(b)	Offset of cement concrete block, which is under masonry sprayed footing, is 15cm, Bearing pressure on soil is 98.75 kN/m^2 and modulus of rapture is 245 kN/m^2. Calculate depth of concrete block.	02
	(c)	State the causes of failure of foundation.	02
Q - 5	(a)	List types of floors. Draw neat sketch of one way and two way R.C.C. slab as a floor.	07
	(b)	Draw neat sketch of Steel roof truss.	05
	(c)	List roof covering material commonly use for pitched roof.	02

OR

Q – 5	(a)	Draw neat Figure of Stone slab floor with and without steel joist.	07
	(b)	State requirement of good roof. Define following terminology with respect to Roof and roofing. a) Span b) Rise c) Ridge d) Pitch e) Common Rafter f) Purlin	07
Q – 6	(a)	State classification of deep foundations. Explain step wise method of driving Simplex pile in to soil. Draw neat figure of Simplex Pile with various stages of driving.	06
	(b)	Write the method of two coat cement plastering.	04
	(c)	Draw any type of underpinning and write its salient points.	04

OR

Q – 6 (a) A brick wall 30 cm thick having height 3.2 m high above plinth level. Plinth level of structure is at 0.45 m from ground level having same thickness of wall. Wall carries superimpose load of 100 kN/m. Unit weight of soil 16.5 kN/m^3, Angle of repose 30^0 and q_s is 158 kN/m^2. Foundation of wall have cement concrete base, which has unit weight of 23.5 kN/m^3, modulus of rapture is equal 245 kN/m^2. Consider unit weight of masonry 19.5 kN/m^3. **06**

Using given data calculate following parameters for strip footing of wall.

a) Pressure at base of footing

b) Width of footing

c) Depth of footing as per Rankines formula

d) Depth of footing as per uniform angle of sprayed from masonry.

(b) What is pointing? Draw figure of Rubbed pointing, Beaded pointing and recessed pointing. **04**

(c) What is underpinning? State purpose of under pinning. **04**

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